Computer Network

Data Communication

❖ Data Communication

- **Data means -** Data is a collection of raw facts, figures, symbols or information that can be processed, stored, and transmitted from one device to another.
  Example : text, numbers, images, audio, video, files etc
- **Communication means** – exchange (sending and receiving) of information.

Q. What is Data Communication ?

Data communication is the process of sending and receiving data from one device to another.

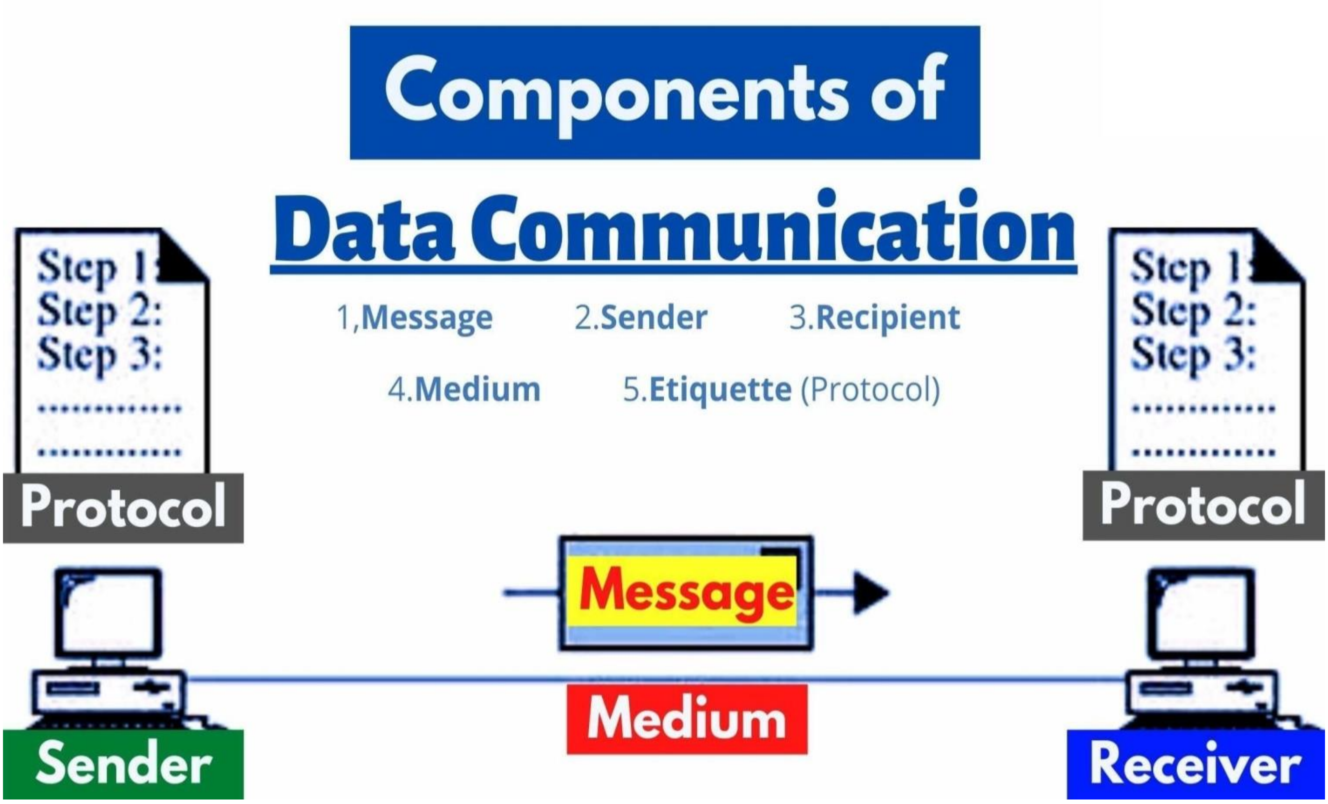
- Data is transferred from one place to another in the form of signals (digital or analog).
- Data is transmitted through mediums like cables, wireless signals or fiber optics.

❖ Characteristics of Data Communication
(Data संचार की विशेषताएँ)

1. **Delivery:** The data should be delivered to the correct destination and correct user.
2. **Accuracy:** The communication system should deliver the data accurately, without introducing any errors. The data may get corrupted during transmission affecting the accuracy of the delivered data.
3. **Timeliness:** Audio and Video data has to be delivered in a timely manner without any delay; such a data delivery is called real time transmission of data.
4. **Jitter(घबराना):** It is the variation in the packet arrival time. Uneven Jitter may affect the timeliness of data being transmitted.

❖ Components of Data Communication

1. **Message :-** Message is the information to be communicated by the sender to the receiver.
2. **Sender :-** The sender is any device that is capable of sending the data (message).
3. **Receiver :-** The receiver is a device that the sender wants to communicate the data (message).
4. **Transmission Medium :-** It is the path by which the message travels from sender to receiver. It can be wired or wireless and many subtypes in both.
5. **Protocol :-** A protocol is a set of rules that data communication. It is an agreed upon set or rules used by the sender and receiver to communicate data.



❖ Types of Data Communication

Data Communication

- **Based on Signals**
  - Digital Signals
  - Analog Signals
  - Hybride Signals
- **Based on Communication Mode/ Channel / data flow (default)**
  - Simplex Channel/Mode
  - Duplex Channel /Mode
    - Half Duplex
    - Full Duplex
- **Based on Media**
  - Guide /Wird Media
  - Unguide/Wirless

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Based on Transmission Mode

(i) **Serial transmission** (ii) **Parallel Transmission**

Note1: "If an exam asks 'Types of Data Communication' without mentioning any basis, answer with the Communication Modes: Simplex, Half-Duplex, and Full-Duplex."

Note2 : Communication mode (data किस दिशा में जाता हैं)

Transmission Mode (data भेजने के तरीके)

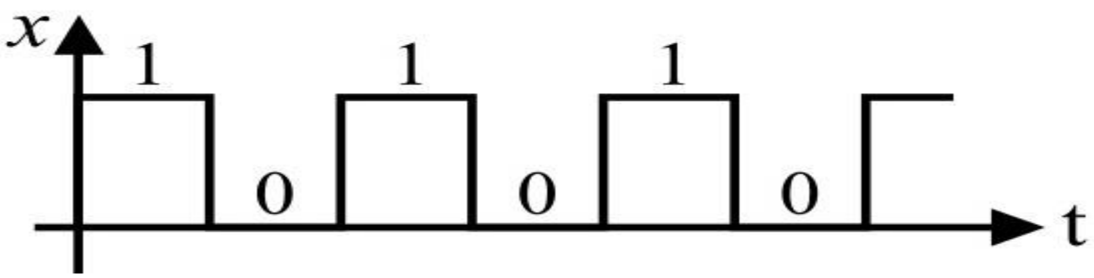
Transmission Media (माध्यम/रास्ता के आधार पर)

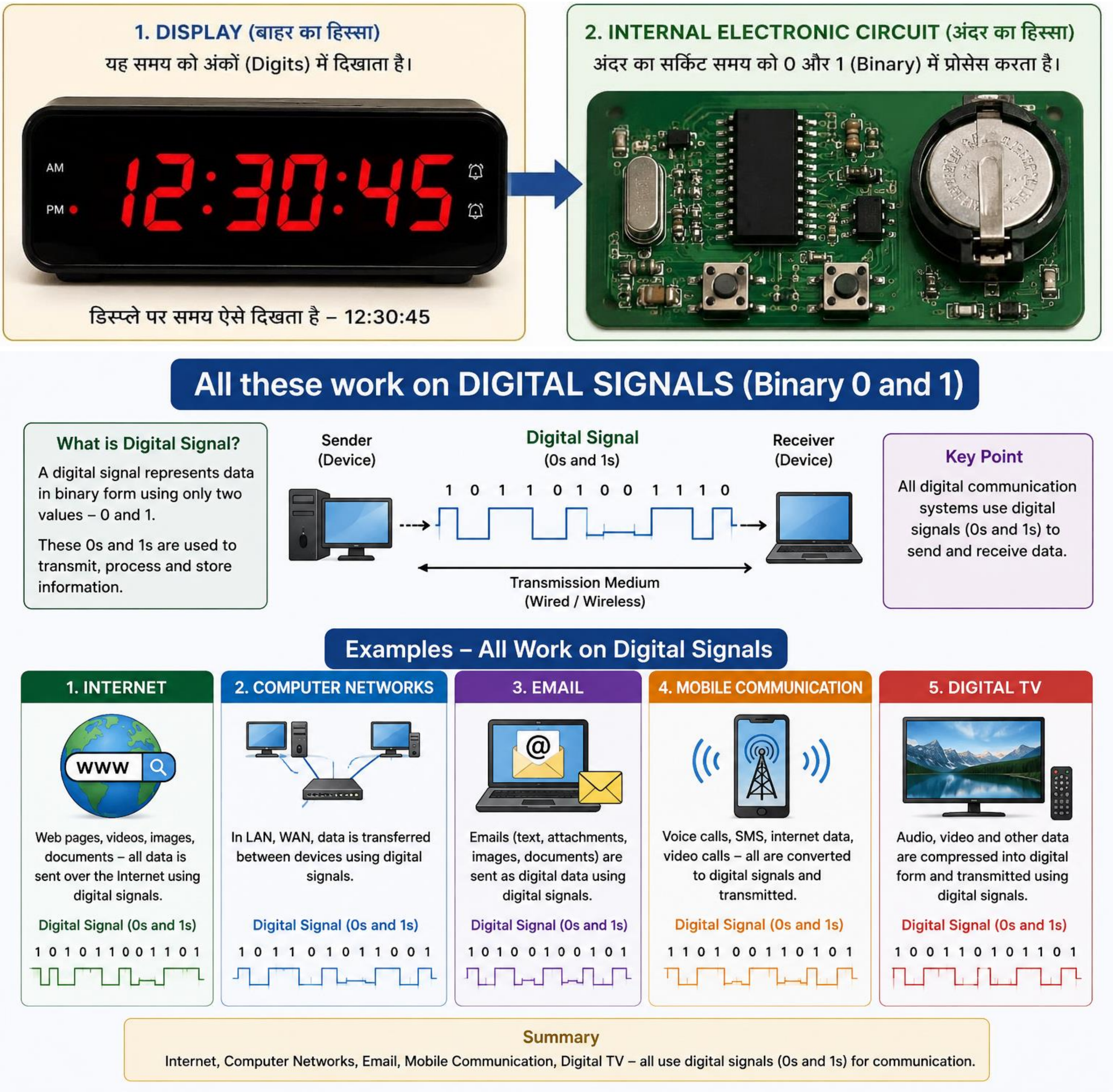
❖ There are three types of signals

1. Digital Signals / Digital Communication

- In digital signals, data/ information is exchanged b/w devices in electronic form, that is, as binary digits (0 and 1).
- Use discrete signals and graphically represent by square wave.
- Example – Internet, Computer network, email, digital tv, mobile communication etc………

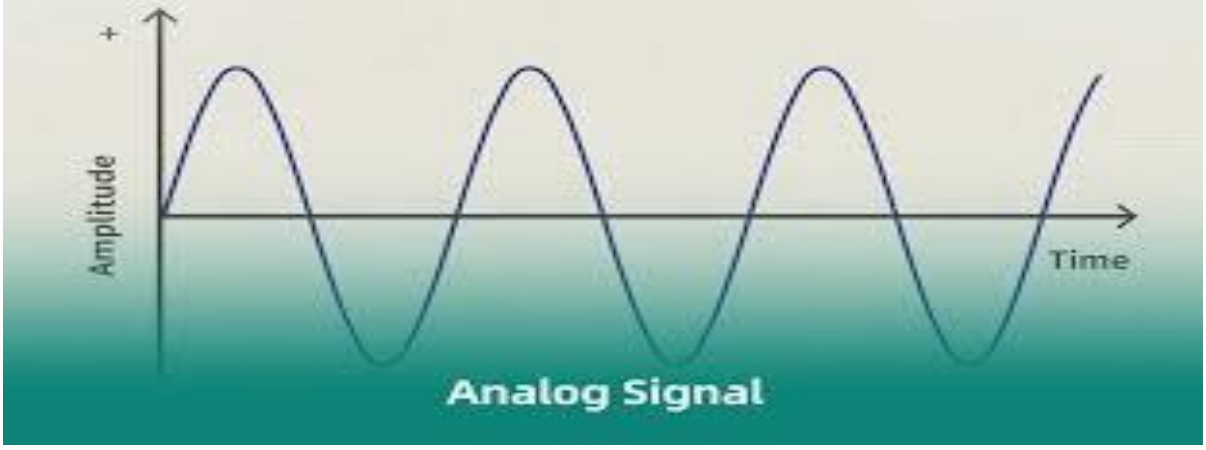
Note : digital clock is not digital communication device, it is only works on digital signal.





2. Analog Signals / Analog Communication

- In analog signals, data /information is exchanged b/w devices in the form of radio waves / waves form.
- Use continuous signals and graphically represented by the sine wave/ continuous wave.
- Example – Traditional landline telephone, AM radio broadcasting, FM radio broadcasting, Analog TV (old TV), walkie-talkie (analog versions)



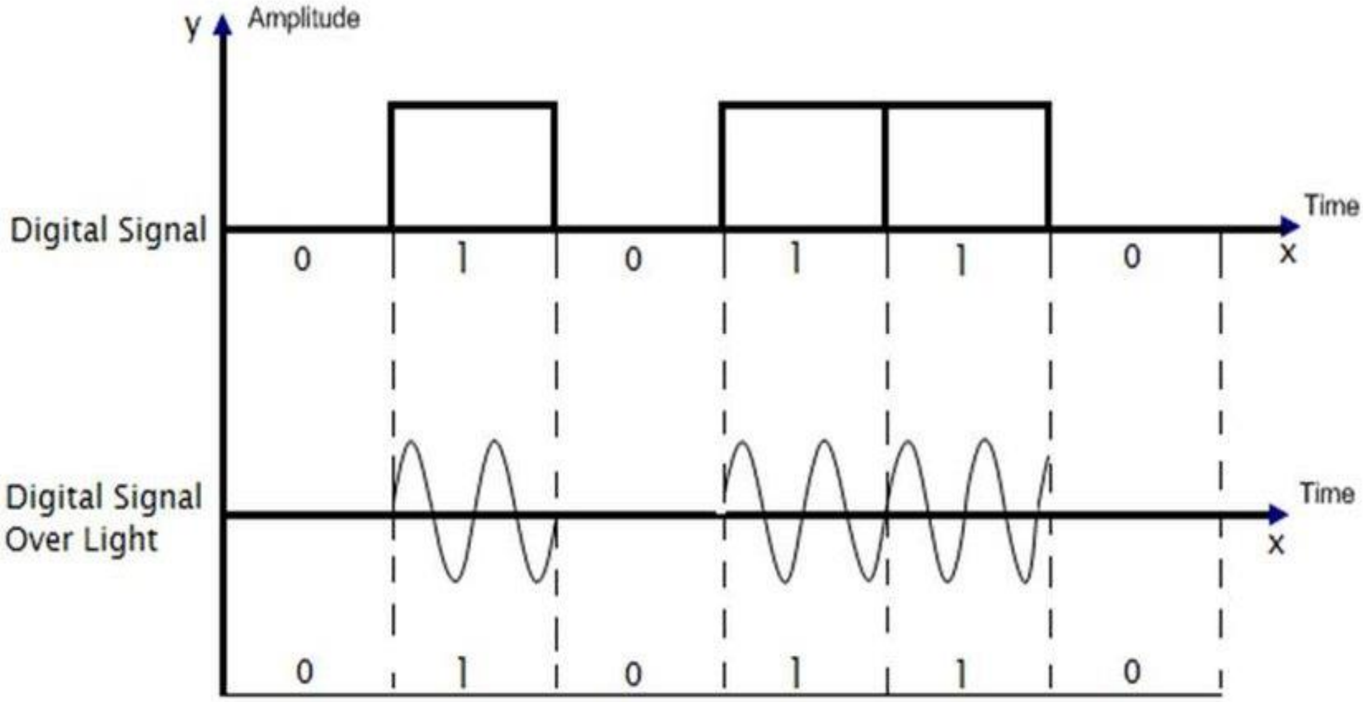
3. Hybrid Signals

- Hybrid signals have properties of both analog and digital signals.
- **ADC (analog to digital converter)** : A method/technique in which analog signal is converted into digital signal.
  Technique : PCM(Pulse code modulation), DM(Delta Modulation,DPCM (Differential PCM)…..
  Devices : Microphone, Digital Camera, Scanner, ADC IC,….

- **DAC (digital to analog converter)** : A method in which digital signal is converted into analog signal.

  Technique :PWM (Pulse Width Modulation),Binary Weighted Resistor DAC,.....

  Device : Speaker, Headphones, Sound Card (Audio DAC), DAC IC
- Example : Digital telephone systems/mobile phone, Audio CDs, internet calling(whatsapp,skype,meet,...), Digital set-up box,



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Extra Topic (Not in Syllabus)

Q. How PCM work ?

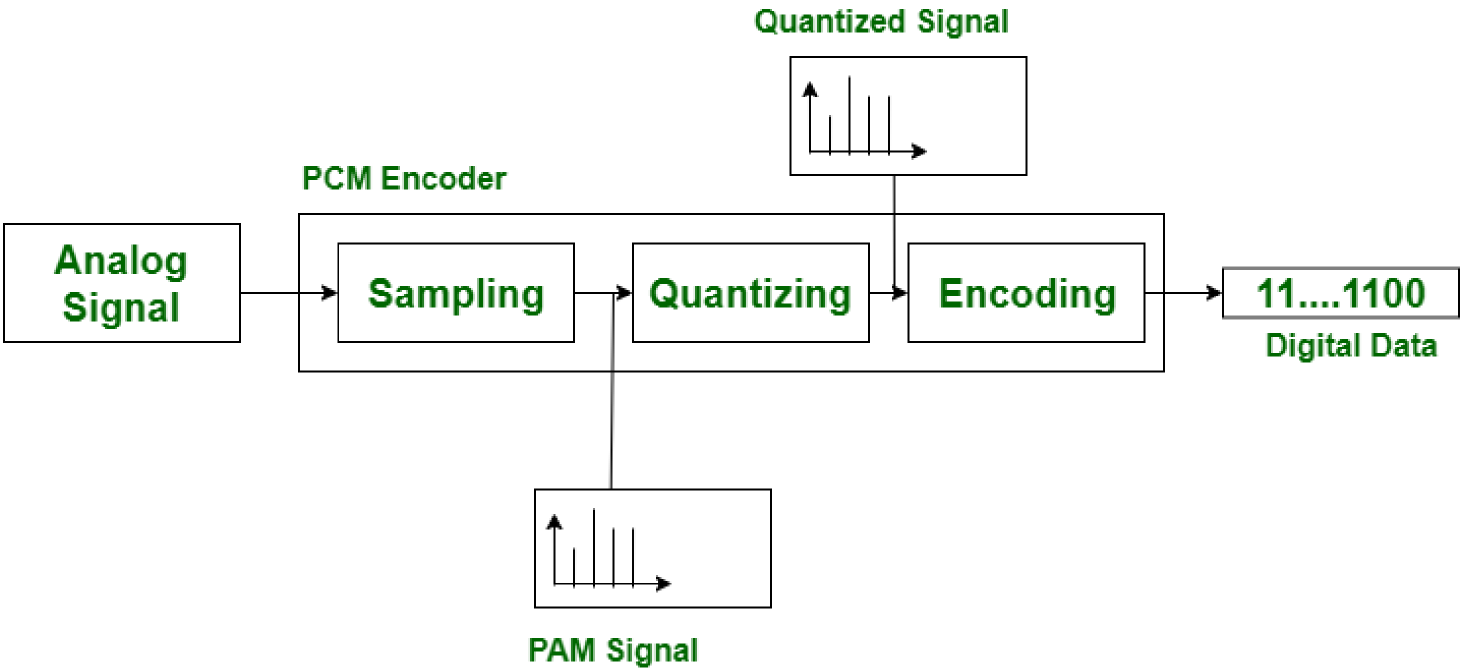
Steps of PCM

(i) Sampling : The analog signal is measured at regular time intervals.

(ii) Quantization : Each sampled value is assigned the nearest fixed level.

(iii) Encoding : Each quantized value is converted into binary (0 and 1)

Q. Block diagram of PCM ?

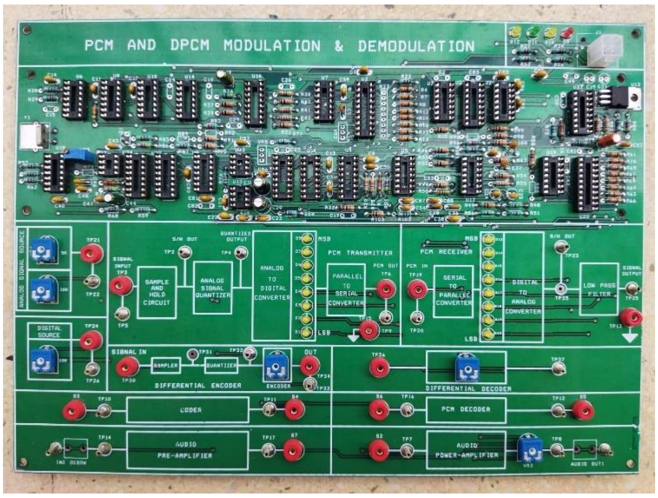


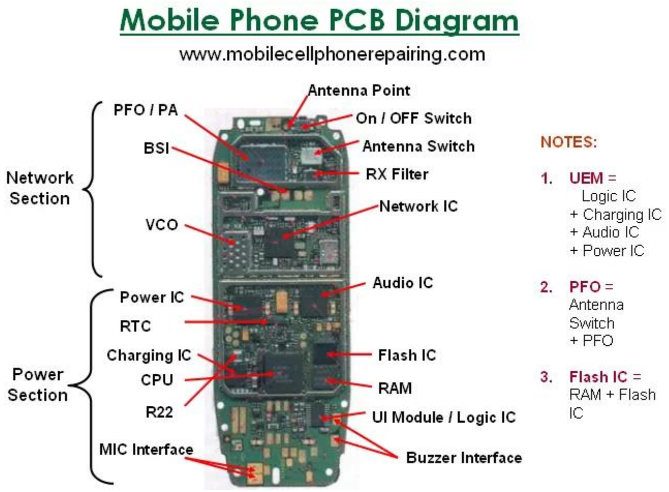
Block Diagram Of PCM

Practical :

Video 1: https://www.youtube.com/shorts/cwsCFOu4OB4

Video 2 : https://www.youtube.com/shorts/cwsCFOu4OB4





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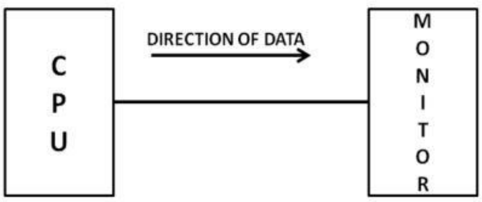
Difference Between Analog & Digital Signals

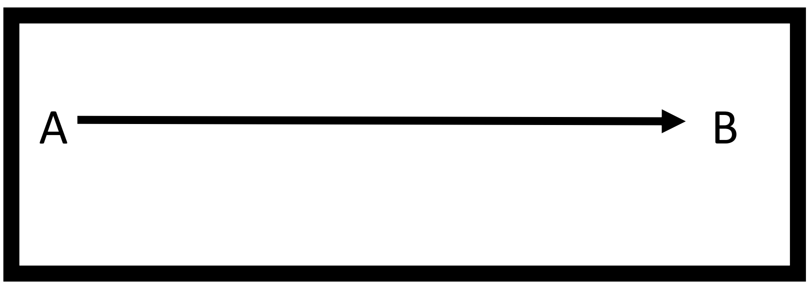
Analog	Digital
A continuous wave that represents data.	A discrete signal that represents data in binary form (0s and 1s).
Requires less bandwidth for transmission.	Requires more bandwidth due to discrete data encoding.
Less expensive to produce and transmit.	More expensive due to advanced technology and processing requirements.
Less flexible. harder to modify or enhance.	Highly flexible. easy to modify, compress, or enhance.

❖ Types of communication channels/Mode

1. Simplex Channel/mode

- In this, the communication is **Unidirectional**, it means data flow in one direction only.
- This channel can transmit data in only one direction.
- Through this channel, only one transmitting device can send information and the other transmitting device can only receive information.
- For example- 1. Radio station from reach the listeners, but do not go back from the listeners to the radio station.
  2. CPU to Monitor-A cpu send data while a monitor only receives data.
  3. Remote
  4. TV broadcasting





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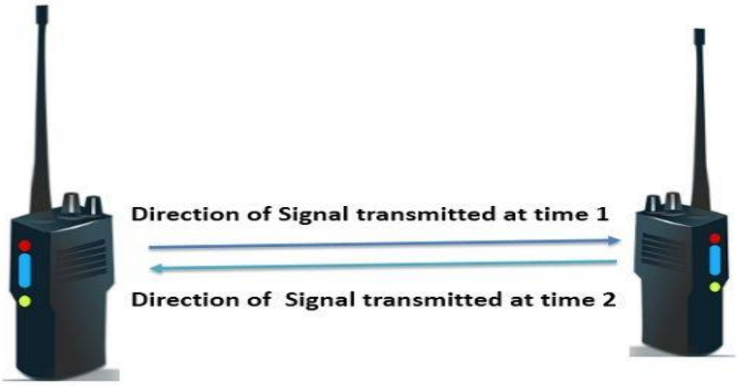
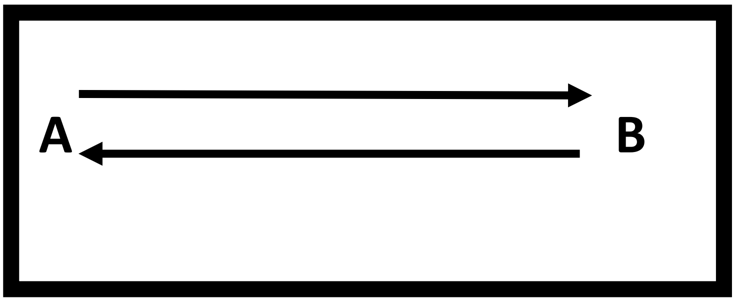
2. Half Duplex Channel / mode

- In this channel, data flows in both directions, but data can flow in only one direction at a time.
- When one device is sending other can only receive and viceversa
- For example, in old telephone line, data is transmitted in only one direction at a time, A walkie-talkie, two-way radios,.

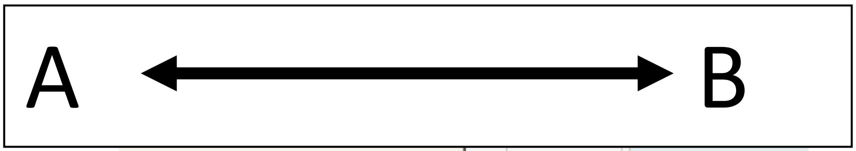


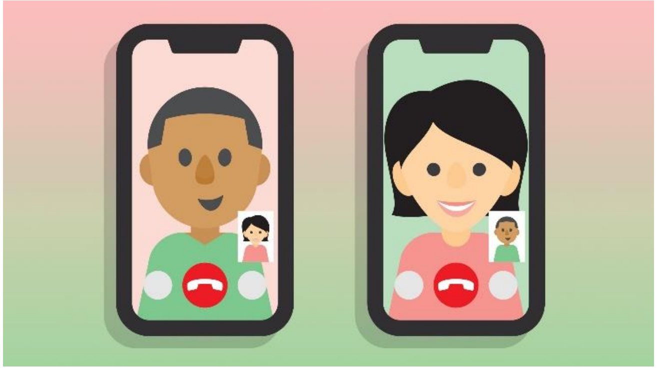


In this mode, two people talk by pressing the PTT (Push-to-Talk) button.
Practicle :
https://www.youtube.com/shorts/LJdAKnafe88?si=Tk8pCf1OAwNJBnyW

3. Full Duplex Channel / mode

- In Full duplex mode, flows data on both side (Sender & Reciver) can transmit and receive at the same time / simultaneously.
- Example: mobile phones, video call, internet communication, VoiP (Voice over internet protocol),.....





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